

Comparative seasonal acclimatization of food and energy consumption in adjacent populations of common spiny mice (*Acomys cahirinus*)

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Abstract

Animals inhabiting environments with low productivity and food availability commonly have reduced energy demands and increased digestive efficiencies. The dry matter intake (DMI), apparent digestible dry matter (ADDM), digestible efficiency (DE) and digestible energy intake (DEI) of two populations of common spiny mouse *Acomys cahirinus* were compared during both winter and summer under conditions of simulated water stress. Mice were captured from the north- and south-facing slopes (NFS and SFS) of the same canyon that represent mesic and xeric habitats, respectively. Measured variables were also compared between F₁ mice that had been born to either NFS or SFS mice, and raised in the laboratory. SFS mice were able to assimilate energy more efficiently than NFS mice during the summer. By comparison, NFS mice were able to assimilate more energy during the winter. During winter, NFS mice assimilated more energy at low levels of water stress, whereas SFS mice assimilated more energy at higher levels. Differences were also apparent in F₁ mice. It is therefore suggested that local climatic conditions can impose physiological adaptations that are retained in succeeding generations, creating unique meta-populations.

Key words: *Acomys*, energy, metabolism, thermoregulation, digestibility

INTRODUCTION

An important aspect that might limit survival in any given habitat is the rate and efficiency at which energy can be acquired. Several environmental factors, such as food availability and food quality, are potentially of importance (Nagy, 1983; Corp, Gorman & Speakman, 1997). Animals living in ecosystems with reduced food availability, such as deserts, commonly have reduced energy demands and an increased efficiency of nutrient uptake, compared with animals from more productive environments (McNab, 1989; Haim, 1987; Mueller & Diamond, 2001). Within Mediterranean ecosystems, different ecological conditions can exist in localized area due to differences in exposure to solar radiation. Environments similar to 'xeric savanna' and 'temperate mesic' can be formed on south- and north-facing slopes (SFS and NFS), respectively, of valleys and canyons due to increased amounts of incident solar radiation (up to 300%) received on the SFS (Nevo, 1995). This striking difference between slopes offers a situation where closely located populations of the same species can inhabit very different habitats. The most well-known locality in recent years is the lower Nahal Oren, the

'Evolution Canyon' (Nevo, 1995). Apart from differences in habitat due to solar radiation, seasonal differences also impose different challenges to the species inhabiting either site. During the summer, the main environmental stressors are likely to be aridity and heat, which present more of a challenge to individuals that inhabit the SFS than the NFS. In addition, as the dry summer season is long (sometimes 8 months), the vegetation is expected to have an increased particle (or osmotic) concentration due to the higher evaporation of water (Grammatikopoulos, 1999). By comparison, during the winter, the main environmental stressor is presumably low ambient temperature, which is expected to affect individuals that inhabit the NFS more than those that inhabit the SFS.

The common spiny mouse *Acomys cahirinus* inhabits both the NFS and the SFS of the lower Nahal Oren (Blaustein, Kotler & Nevo, 1996). We have previously shown that there are a number of physiological differences between *A. cahirinus* originating from these two areas. For example, NFS mice have higher field metabolic rates and a greater capacity to produce heat by non-shivering thermogenesis (NST) than SFS mice (Scantlebury, Afik *et al.*, 2002; Scantlebury, Shanas, Speakman *et al.*, 2003). In the current study, we aim to examine whether populations of *A. cahirinus* originating from the NFS and SFS of Nahal Oren also differ in their seasonal ability to acquire and process energy. Also, to examine whether

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the two populations differ when exposed to water stress (reflecting the poorer ability of the NFS mice to cope with the increased aridity towards the end of the dry season) (Shanas *et al.*, 2002); and whether energetic differences are evident in F_1 mice, which may indicate genetic as well as phenotypic components to any variation.

MATERIALS AND METHODS

Animals and housing

Acomys cahirinus were trapped from the NFS and SFS of the lower Nahal Oren, Mount Carmel, Israel (32°43'N, 34°58'E) and taken to the Department of Biology at the University of Haifa-Oranim during the summer (July–August, 1999). Animals ('founders') were placed individually in cages (35 × 25 × 15 cm) that were kept in an open shaded area, which did not receive direct sunlight or rain but was otherwise exposed to outdoor conditions. They were kept throughout the summer (c. 14L:10D, min/max 28°C/37°C, July–August 1999) and the following winter (c. 10L:14D, min/max 10°C/21°C, January–February 2000). There were 2 experimental periods (summer and the following winter) for founder-mice, each lasting a total of 9 weeks. Between these periods, mice were maintained on an *ad libitum* diet of carrots and rodent chow (Koffolk, Israel, 21% crude protein, 4% crude fat, 4% cellulose, 13% moisture, 7% ash, 18.7 kJg⁻¹ gross energy). Following the winter session, NFS and SFS mice were housed in 2 separate colonies and allowed to breed. The same experiments were carried out on F_1 mice in a 9-week period the following winter (January–February 2001), when they were c. 1 year old (Shanas *et al.*, 2003). Laboratory paper towelling was provided as bedding/nesting material during the experimental periods (4 sheets of paper of c. 30 × 40 cm per sheet), which was available to animals at all times. In between experimental periods, spiny mice were provided with fresh sawdust and paper towelling. Spiny mice were regularly examined by a veterinarian and all experimental procedures and animal husbandry practices were approved by the Animal Ethics Committee, University of Haifa. On completion of experiments, animals were maintained in the laboratory to form a breeding colony.

Food intake and digestibility measurements

During acclimation periods (see below), mice were offered dry soybeans *ad libitum* (17.4 kJ/g gross energy) and agar gel as a source of moisture (20 g of dry agar agar dissolved in 1000 ml of water). After acclimation to each level of salinity, cages were cleaned and each mouse was offered a known mass of dry soybeans (c. 60 g). One week later, the remaining soybeans and the faeces produced were removed from each cage and dried to constant weight in an oven (Mettler, Germany; 60°C). Subsequently, for each mouse, at each level of salinity, the dry food and faeces remains were separated by hand, weighed

(Sartorius, Germany; ± 0.01 g), homogenized, and the faecal calorific value (FE, kJ/g) was determined by adiabatic bomb calorimetry (semimicro calorimeter: 1425. Parr, U.S.A.). Hence, for each mouse, at each level of salinity, measures of food intake and faecal output were obtained over the period of a week. Dry matter intake (DMI, g) was measured as the mean dry food consumed per day and faecal output (FO, g) as the mean dry faeces produced per day. Apparent digestible dry matter (ADDM) was calculated as:

$$\text{ADDM} = (\text{DMI} - \text{FO}) \quad (1)$$

Apparent digestible energy intake (DEI) was calculated as:

$$\text{DEI (kJ/day)} = [\text{DMI (g)} \times 17.4 \text{ (kJ/g)}] - [\text{FO (g)} \times \text{FE (kJ/g)}] \quad (2)$$

Digestible efficiency (DE) (%) was calculated as:

$$\text{DE} = [(\text{DMI} - \text{FO})/\text{DMI}] \times 100 \quad (3)$$

(Haim, Van Aarde & Skinner, 1990).

Salt intake protocol

To simulate changes of osmotic load experienced in the wild, animals were exposed to increasing levels of water stress by dissolving known quantities of NaCl into their water source (the agar gel). As the only food provided during experimental periods was protein-rich soybeans (41% protein, Weissenberg & Shkolnik, 1994), mice were forced to drink the salty water to reduce urea concentrations. During the summer, founder mice were acclimated to 0.9%, 2.5% and then 3.5% salinity. During the winter, both founders and F_1 mice were acclimated to 0.9%, 1.8% and then 2.5%. Mice were not acclimated to 3.5% salinity in the winter because we have shown previously that mice were not able to withstand this salinity load during this time (Shanas *et al.*, 2002) (at 3.5% salinity during the winter some mice lost 2–3 g/day (5–10%) of body mass and seemed very poorly after only 2–3 days). Acclimation to each level of salinity lasted 2 weeks (Shanas *et al.*, 2003).

Analyses

Mixed linear models were used to determine whether salinity, site of origin (NFS or SFS), season (winter or summer) or generation (founder or F_1) had an effect on the rates and efficiency of energy acquisition (i.e. DMI, ADDM, DE and DEI) in *A. cahirinus* (Pinhiero & Bates, 2000). Five separate analyses were performed that examined differences in measured variables between: founder mice from different sites of origin during the summer; founder mice from different seasons; founder mice from different slopes of origin during the winter; mice from different generations during the winter; and F_1 mice from different sites of origin during the winter. Separate models for DMI, ADDM, DE and DEI were fitted using

residual maximum likelihood (REML) and statistical significance of fixed effects was inferred using conditional F -tests (Pinheiro & Bates, 2000). Salinity, site of origin, season and generation were fitted as fixed factors, body mass as a fixed covariate and animal identity as a random factor. The most parsimonious model for each dependent variable was selected by sequentially deleting terms from full models using Akaike's information criterion (AIC) (Aikake, 1974). Statistical analyses were carried out using R version 2.0.0 for Linux (<http://www.r-project.org>). Sample sizes varied slightly between treatments because 1 individual died on acclimation to 2.5% salinity, and a further 4 individuals died between experimental periods from summer to winter. Sample sizes also differed slightly in the F_1 mice because of loss of 1 of the samples during analysis.

RESULTS

Effect of site of origin during the summer in founders

Values of DE were greater in SFS mice than NFS mice ($F_{1,22} = 7.19$, $P = 0.01$) and DE and DEI increased in mice from both sites of origin with increasing salinity ($F_{2,15} = 31.21$, $P < 0.01$ and $F_{2,13} = 6.52$, $P = 0.01$, respectively) (Table 1a). There was no difference in DMI or ADDM between NFS and SFS mice ($F_{1,22} = 2.78$, $P = 0.11$ and $F_{1,22} = 0.50$, $P = 0.40$, respectively).

Effect of season in founders

DE was significantly greater in the winter than in the summer ($F_{1,25} = 17.6$, $P < 0.01$). There were significant three-way interactions between season, salinity and site of origin for DMI, ADDM and DEI ($F_{1,15} = 7.34$, $P = 0.02$; $F_{1,15} = 8.75$, $P = 0.01$; $F_{1,8} = 8.08$, $P = 0.02$, respectively) (Table 1a,b): During the winter, values were generally higher in NFS mice at low levels of salinity but greater in SFS mice at higher levels of salinity. During summer, values increased in both NFS and SFS mice but were always greater in NFS mice.

Effect of site of origin during the winter in founders

There were significant interactions between site of origin and salinity for DMI, ADDM and DEI ($F_{2,13} = 7.90$, $P = 0.01$; $F_{2,13} = 6.12$, $P = 0.01$; $F_{2,8} = 9.39$, $P = 0.01$, respectively) (Table 1b): NFS mice had greater values at low levels of salinity, whereas SFS mice had greater values at higher levels of salinity. There was no difference in DE between NFS and SFS mice ($F_{1,9} = 1.77$, $P = 0.22$).

Effect of generation during the winter

There were significant interactions between salinity and generation for DMI, ADDM, DE and DEI ($F_{2,40} = 3.70$, $P = 0.03$; $F_{2,40} = 3.93$, $P = 0.03$; $F_{2,40} = 7.52$, $P < 0.01$;

$F_{2,27} = 3.48$, $P = 0.05$) (Table 1b,c); values were greater in founder mice at low levels of salinity but greater in F_1 mice at higher levels of salinity. There were also significant interactions between site of origin and salinity for DMI, ADDM and DEI ($F_{2,40} = 5.34$, $P = 0.01$; $F_{2,40} = 4.64$, $P = 0.02$; $F_{2,27} = 4.27$, $P = 0.02$, respectively); values were greater in NFS mice at low levels of salinity but greater in SFS mice at higher levels of salinity.

Effect of site of origin in F_1 mice during the winter

DMI and DEI were significantly greater in SFS mice than NFS mice ($F_{1,17} = 5.61$, $P = 0.03$; $F_{1,17} = 4.49$, $P = 0.05$) (Table 1c). DMI, ADDM, DE and DEI were significantly different across salinity levels ($F_{2,26} = 31.0$, $P < 0.01$; $F_{2,26} = 33.0$, $P < 0.01$; $F_{2,24} = 36.0$, $P < 0.01$; $F_{2,24} = 34.0$, $P < 0.01$, respectively).

DISCUSSION

The ability of desert rodents to survive on sparse, unpredictable, poor quality food together with water shortages is well documented (Schmidt-Nielsen, 1964; Louw & Seely, 1982; Degen *et al.*, 2000). In general, they have lower absolute energy demands, lower resting metabolic rate (RMR) values and more efficient digestive systems compared with similarly sized rodents inhabiting mesic areas (McNab, 1980, 1983; Nagy & Gruchacz, 1994). For example, the Saharo-Arabian desert-living fat jird *Meriones crassus* is able to survive on low quality roughage by keeping energy requirements to a minimum whilst maintaining a high DE. By comparison, the Levant vole *Microtus guentieri*, a mesic Mediterranean-living species, requires approximately twice the DEI of *M. crassus*, and is unable to maintain a high DE when offered low quality roughage (Choshniak & Yahav, 1987). Differences in the rates and efficiencies of energy acquisition also occur within species from populations inhabiting different environments. Studies on the degu *Octodon degus*, a small herbivorous rodent that inhabits both arid and semi-arid Mediterranean regions of Chile, have shown that increased dietary fibre results in increased food intake and faecal output but decreased metabolic rates and DE (Veloso & Bozinovic, 1993; Bozinovic, 1995). The authors suggested that the populations of degus were able to adjust physiologically to changes in the environment, in particular, by reducing metabolic demands as a response to a decreasing nutrient and energy availability. Similarly, *A. cahirinus* have been shown to exhibit significantly higher values in all food and energy consumption variables when under short photoperiod acclimatization compared with long photoperiod acclimatization (Haim, Afik & Shanas, 2001).

In the current study, populations of *A. cahirinus*, which inhabit adjacent microhabitats on the opposite slopes of Nahal Oren, exhibit seasonal differences in their capacities to digest food and acquire energy. Consistent with our expectations, summer values of DE were higher in mice

Table 1. Mean and standard deviations (SD) of body mass (mass; g), dry matter intake (DMI; g/day and g/Wb.day), faecal output (FO; g/day and g/Wb.day), apparent digestible dry matter (ADDM; g/day and g/Wb.day), digestible efficiency (DE; %) and digestible energy intake (DEI, kJ/day and kJ/Wb.day) for north facing slope (NFS) and south facing slope (SFS) *Acomys cahirinus* measured in: (a) the founder population during the summer (Summer); (b) the founder population during the winter (Winter); (c) F₁ offspring during the winter (F₁-winter). Salinity regimes were 0.9%, 2.5% and 3.5% during the summer and 0.9%, 1.8% and 2.5% during the winter. *n*, sample sizes; Wb, body mass (g) of the animal

	NFS			SFS		
(a) Summer						
Salinity	0.9	2.5	3.5	0.9	2.5	3.5
<i>n</i>	8	7	7	7	7	7
mass (g)	46.36	45.04	41.54	44.16	40.40	35.27
SD	5.78	5.10	7.74	5.60	7.18	6.43
DMI (g/day)	2.00	2.08	2.05	1.55	1.95	1.71
SD	0.32	0.39	0.53	0.62	0.50	0.29
DMI (g/Wb.day)	0.044	0.047	0.050	0.035	0.049	0.049
SD	0.007	0.010	0.013	0.012	0.016	0.008
FO (g/day)	0.357	0.364	0.243	0.182	0.317	0.152
SD	0.181	0.083	0.049	0.054	0.830	0.057
FO (g/Wb.day)	0.0076	0.0081	0.0060	0.0041	0.0078	0.0045
SD	0.0036	0.0018	0.0015	0.0010	0.0019	0.0014
ADDM	1.58	1.71	1.79	1.36	1.63	1.56
SD	0.18	0.33	0.52	0.57	0.43	0.25
ADDM (g/Wb.day)	0.034	0.039	0.044	0.031	0.041	0.045
SD	0.005	0.009	0.013	0.011	0.014	0.007
DE (%)	82.3	82.4	87.6	87.9	83.6	90.9
SD	7.3	2.2	3.1	1.7	2.1	2.5
DEI (kJ/day)	32.32	26.33	35.76	26.90	24.92	30.37
SD	3.76	5.13	9.80	10.99	6.44	5.03
DEI (kJ/Wb.day)	0.81	0.65	0.94	0.65	0.68	0.92
SD	0.14	0.15	0.25	0.23	0.22	0.14
(b) Winter						
Salinity	0.9	1.8	2.5	0.9	1.8	2.5
<i>n</i>	5	5	5	5	5	5
mass (g)	51.70	51.56	50.68	63.14	59.30	55.65
SD	6.18	4.77	4.78	10.88	4.00	3.78
DMI (g/day)	3.72	4.91	2.93	3.31	4.97	3.81
SD	0.76	0.54	0.15	0.39	0.48	0.27
DMI (g/Wb.day)	0.073	0.095	0.058	0.054	0.084	0.069
SD	0.016	0.009	0.006	0.013	0.008	0.005
FO (g/day)	0.439	0.431	0.301	0.382	0.569	0.415
SD	0.146	0.096	0.036	0.063	0.136	0.036
FO (g/Wb.day)	0.0088	0.0083	0.0059	0.0060	0.0100	0.0070
SD	0.0035	0.0016	0.0005	0.0016	0.0002	0.0007
ADDM (g/day)	3.28	4.48	2.63	2.91	4.42	3.39
SD	0.69	0.47	0.13	0.38	0.40	0.25
ADDM (g/Wb.day)	0.064	0.087	0.052	0.048	0.075	0.061
SD	0.014	0.008	0.006	0.012	0.007	0.005
DE (%)	88.2	91.3	89.7	88.0	89.0	89.1
SD	3.4	1.3	1.1	2.1	1.9	0.5
DEI (kJ/day)	65.11	87.25	51.81	58.05	87.26	67.01
SD	13.41	9.20	2.71	7.18	8.20	4.85
DEI (kJ/Wb.day)	1.27	1.70	1.03	0.96	1.47	1.21
SD	0.28	1.54	0.11	0.24	0.13	0.10
(c) F₁-winter						
Salinity	0.9	1.8	2.5	0.9	1.8	2.5
<i>n</i>	8	7	7	8	9	9
mass (g)	42.91	46.15	44.91	49.05	53.10	51.83
SD	6.68	5.19	5.33	8.52	5.07	4.66
DMI (g/day)	3.23	4.46	3.42	3.25	4.81	4.09
SD	0.26	0.75	0.51	0.38	0.62	0.34
DMI (g/Wb.day)	0.077	0.098	0.077	0.068	0.092	0.079
SD	0.013	0.022	0.012	0.015	0.016	0.009
FO (g/day)	0.494	0.377	0.372	0.494	0.514	0.445
SD	0.067	0.073	0.065	0.087	0.110	0.058
FO (g/Wb.day)	0.0117	0.0816	0.0084	0.0102	0.0096	0.0087
SD	0.0022	0.0013	0.0018	0.0019	0.0042	0.0016

Table 1. Continued.

NFS				SFS		
(c) F ₁ -winter (cont.)						
ADDM	2.73	4.07	3.04	2.75	4.29	3.65
SD	0.21	0.75	0.52	0.35	0.64	0.34
ADDM (g/Wb.day)	0.065	0.090	0.068	0.058	0.082	0.071
SD	0.011	0.022	0.011	0.014	0.017	0.009
DE (%)	84.7	91.4	88.9	84.7	89.2	89.1
SD	1.4	2.2	2.5	2.4	2.5	17.0
DEI (kJ/day)	55.01	79.50	60.20	55.49	84.60	71.78
SD	4.27	14.10	9.47	6.74	11.47	6.22
DEI (kJ/Wb.day)	1.31	1.75	1.35	1.16	1.61	1.40
SD	0.22	0.41	0.21	0.26	0.32	0.16

from the SFS and increased more rapidly in response to dietary salinity than values in mice from the NFS. This indicates that, when fed the same diet, but under water stress, SFS mice were able to extract more energy from foodstuffs than NFS mice. Thus, SFS mice may be able to cope with poorer quality diets (containing less water and more fibre) during the summer than NFS mice. During winter, ambient temperatures were *c.* 15 °C lower than those measured during the summer (Shanas *et al.*, 2002) and all mice consumed roughly twice the amount of food (DMI), which they digested with greater efficiency. This extra energy was probably needed to cover the increased costs of thermoregulation (Kronfeld *et al.*, 1998). In addition, NFS mice had greater DEI values at low levels of salinity whereas SFS mice had greater DEI values at higher levels of salinity. This indicates that SFS and NFS mice respond differently across different seasons; SFS mice can assimilate energy more effectively during summer under water stress, whereas, NFS mice can assimilate energy more effectively during the winter, under wet and cold conditions. This agrees with earlier work, which shows that resting metabolic rate values decreased more rapidly in response to increasing salinity in NFS mice (Scantlebury, Afik *et al.*, 2002), and that energy expenditure *per se* was greater on the cooler and wetter NFS (Scantlebury, Shanas, Speakman *et al.*, 2003; Scantlebury, Shanas, Kupshtein *et al.*, 2004).

It is important to note that differences in food and energy consumption between NFS and SFS mice were also evident in the F₁ generation. Our results showed that, in F₁ mice, the DMI ADDM and DEI of SFS individuals at higher levels of salinity were greater than those of the NFS. As these mice had been born and raised in identical conditions, this indicates that in addition to phenotypic plasticity, there may also be a genetic component to observed variation between the two populations. This is consistent with the suggestion that there are genetic allozyme and RAPD differences between populations of mice from the two slopes (Nevo *et al.*, 1998) and that NFS F₁ mice have greater thermoregulatory capabilities than SFS F₁ mice (Scantlebury, Shanas, Kupshtein *et al.*, 2003). In the current study, there also seemed to be physiological differences between founder and F₁ mice: for example, F₁ mice had higher ADDM values and a better ability to assimilate energy when they were exposed to water stress

than founder mice. Reasons for this remain unclear but F₁ mice were heavier than founders (Scantlebury, Shanas, Kupshtein *et al.*, 2003), which could be a result of being born and raised in captivity to mothers kept on *ad libitum* food. Hence, part of the difference may be an increased stomach capacity and associated ability to digest food (Hammond & Wunder, 1991; Derting & Noakes, 1995).

Studies of rodents from different habitats and under different laboratory conditions have shown that gut morphology may be modified in response to feeding habits and energy requirements (Corp *et al.*, 1997; Degen *et al.*, 2000). Increased energy demands or reduction in food quality result in increased gut length and capacity coupled with increased food intake rates (Hammond & Wunder, 1991, 1995; Geluso & Hayes, 1999). Therefore, differences in DE and DEI between mice from different sites and across different seasons are also likely to be the result of changes in gut morphology. Interestingly, whereas NFS mice maintain relatively high RMR values throughout the year, SFS mice decreased their RMR values during the summer (Scantlebury, Shanas, Speakman *et al.*, 2003). Hence, SFS mice seem to be able to reduce their metabolic demands in response to decreased nutrient and energy availability.

In summary, NFS and SFS *A. cahirinus* from the opposite slopes of Nahal Oren behave differently when acclimatized to summer and winter conditions. SFS mice are able to assimilate energy more efficiently than NFS mice when exposed to water stress during the summer. By comparison, NFS mice are able to assimilate more energy during the winter when under minimal water stress. Bearing in mind that *A. cahirinus* originate from Ethiopia (Bogos/Eritrea; Bates, 1994), it is the SFS mice that retain more of the ancestral characteristics in terms of the response to water stress. The physiological plasticity of NFS mice probably enabled them to inhabit the more mesic, colder and wetter NFS.

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